SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

(Deemed to be University)

Department of Mathematics and Computer Science

SYLLABUS - UMAM-602 - Techniques in Applied Mathematics

L – T – P – C	4 – 0 – 0 – 4	**Prerequisites**	ODE	Total Periods: **56**
Course Category	DSC			

Course Objectives

The course is aimed to lay a broad foundation for an understanding of the problems of the calculus of variations and its many methods and techniques and to prepare students for the study of modern optimal control theory. To make the students familiar with the methods of solving Integral Equations. It also familiarizes the Transform techniques to solve some of the fundamental problems well known in the scientific literature.

Course Outcomes

- Students will be able to appreciate variational formulations.
- Students can apply different methods to solve Integral Equations and variational problems.
- Students can identify the difference between Volterra and Fredholm Integral Equations of the First kind and Second kind, homogeneous and inhomogeneous.
- Students can apply the methods to reduce Initial value problems associated with linear differential equations to various integral equations.
- Students will be able to solve problems of algebraic, differential and integral equations using the transform techniques.
- Students can apply the concept of Fourier transforms to the real-world problems of circuit analysis and control system design. Students would be able to appreciate the power of complex analytic techniques in applying transforms.

Syllabus

Unit	Topic	Contents	Periods
I	**Calculus of Variation**	Variational Problems, Necessary Condition for Extrema, The Simplest Problem, Generalizations, Isoperimetric Problems	12
II	**Integral Equations**	Volterra Equations, Fredholm Equations with degenerate kernels, Symmetric Kernels	12

Unit	Topic	Contents	Periods
III	**Fourier Transforms and Their Applications**	The Fourier Integral Formulas, Definition and Examples, Fourier Transforms of Generalized functions, Basic Properties of Fourier Transforms, Poisson's Summation Formula, The Shannon Sampling Theorem, Gibb's Phenomenon, Heisenberg's Uncertainty Principle, Applications of Fourier Transforms to Ordinary Differential Equations, Solutions of Integral Equations, Solutions of Partial Differential Equations, Fourier Cosine and Sine Transforms with Examples, Properties of Fourier Cosine and Sine Transforms, Applications of Fourier Cosine and Sine Transforms to Partial Differential Equations, Evaluation of Definite Integrals, Applications of Fourier Transforms in Mathematical Statistics, Multiple Fourier Transforms and Their Applications	20
IV	**Hankel Transforms and Their Applications**	The Hankel Transform and Examples, Operational Properties of the Hankel Transform, Applications of Hankel Transforms to Partial Differential Equations	12
		Total	**56**

Key Text(s)

- J. David Logan, Applied Mathematics, John Wiley, 3rd Edition, 2006. Ch: 3 (sec 3.1 to 3.6, excluding 3.5), 4 (sec 4.3 only).
- Lokenath Debnath, Dambaru Bhatta, Integral Transforms and Their Applications, 2nd Edition, Chapman and Hall/CRC, 2006. Ch: 2, 7.

Suggested/Additional Readings

- Brian Davies, Integral Transforms and Their Applications, Springer, 2002.